

Supplementary analyses

2025-12-08

Table S1

Using the Macrogastrinae benchmark to estimate ACG and global Braconidae, Ichneumonoidea, and Hymenoptera (following the same approach we have used for Insecta). We have added Table S1 to present these estimates, which we discuss briefly in the Main Text.

```
library(SpadeR)
library(dplyr)
library(stringr)
library(pairwiseCI)
#library(remotes)
#install_version("CARE1", "1.1.0")
library(CARE1)

## Load Data

#Microgastrinae

reared <- read.csv("../Data/Microgastrinae ACG Reared.csv") %>%
  select(BOLD.code, Frequency)

core <- read.csv("../Data/Microgastrinae ACG Core Malaise.csv") %>%
  select(BOLD.code, Frequency)

peri <- read.csv("../Data/Microgastrinae ACG Peripheral Malaise.csv") %>%
  select(BOLD.code, Frequency)

# All insecta core

insec <- read.csv("../Data/Insecta ACG Core Malaise.csv") %>%
```

```

    select(BOLD.code, Frequency)

#Braconidae

Brac <- read.csv("../Data/Braconidae ACG Core Malaise.csv") %>%
  select(BOLD.code= BOLD.Code, Frequency)

#Ichneumonoidea

Ichn <- read.csv("../Data/Ichneumonoidea ACG Core Malaise.csv") %>%
  select(BOLD.code, Frequency)

#Hymenoptera

hym <- read.csv("../Data/Hymenoptera ACG Core Malaise.csv") %>%
  select(BOLD.code= BOLD.Code, Frequency)

## highlight estimator

estimator <- "    iChao1 (Chiu et al. 2014)"

## adding in the trees

#TA = number of tree species in ACG
TA <- 1500
TA_alt <- 1200

#Tc = number of tree species in CR

TC <- 2500
#Tw = number of tree species on Earth

## estimated tree species

TW <- 73274
TW_CI <- c(73271, 73276)

TW_2 <- 89147

## described number of tree species

TW_d <- 60065

```

```
### Number of Saturniid moths
```

```
SA <- 106
```

```
SW <- 4535
```

Approach 1: Calculate insect richness worldwide using the CARE estimator for Microgastrinae

```
### Total number of BINS
```

```
reared %>% nrow()
```

```
[1] 886
```

```
core %>% nrow()
```

```
[1] 388
```

```
peri %>% nrow()
```

```
[1] 576
```

```
###
```

Step 1: Combining Malaise and Reared samples of Microgastrinae and estimating the true richness of ACG Microgastrinae

```
full_micro_table <- reared %>%  
  rename(reared = Frequency) %>%  
  full_join(rename(core, core = Frequency)) %>%  
  full_join(rename(peri, peri = Frequency)) %>%  
  mutate(reared = ifelse(is.na(reared), 0, 1)) %>%  
  mutate(core = ifelse(is.na(core), 0, 1)) %>%  
  mutate(peri = ifelse(is.na(peri), 0, 1))
```

```
Joining with `by = join_by(BOLD.code)`
Joining with `by = join_by(BOLD.code)`
```

```
head(full_micro_table)
```

	BOLD.code	reared	core	peri
1	BOLD:AAA1266	1	0	0
2	BOLD:AAA1634	1	1	1
3	BOLD:AAA2143	1	0	0
4	BOLD:AAA2511	1	0	0
5	BOLD:AAA2668	1	0	0
6	BOLD:AAA3229	1	0	1

```
nrow(full_micro_table)
```

```
[1] 1411
```

```
# Transform observed ../Data to CARE1 input format
micro_care <- as.record(full_micro_table[, -1])
```

```
micro_care
```

```
001 010 011 100 101 110 111
246 133 146 663 114 39 70
```

```
# Main step to obtain population size estimates
CARE1.print(micro_care)
```

```
Warning in ni/nhat: Recycling array of length 1 in vector-array arithmetic is deprecated.
Use c() or as.vector() instead.
```

```
Warning in ni/nhat: Recycling array of length 1 in vector-array arithmetic is deprecated.
Use c() or as.vector() instead.
```

```
Warning in ni/nhat: Recycling array of length 1 in vector-array arithmetic is deprecated.
Use c() or as.vector() instead.
```

```
(1) NUMBER OF IDENTIFIED CASES IN EACH LIST:
```

```
  n1  n2  n3
886 388 576
```

(2) ESTIMATES BASED ON ANY PAIR OF SAMPLES:

	Petersen	Chapman	se	cil	ciu
pa12	3154	3136	236	2725	3655
pa13	2774	2765	149	2501	3087
pa23	1035	1033	37	970	1115

Note1: Refer to Seber(1982,pages 59 and 60) for Petersen estimator and Chapman estimators as well as s.e formula.

Note2: A log-transformation is used is used to obtain the confidence interval so that the lower limit is always greater than the number of ascertained. Refer to Chao(1987,Biometrics,43,783-791) for the construction of the confidence interval.

(3) SAMPLE COVERAGE APPROACH:

	M	D	Chat	est	se	cil	ciu
Nhat-0	1411	1063.667	0.494	2153	61	2043	2283
Nhat	1411	1063.667	0.494	3480	529	2675	4799
Nhat-1	1411	1063.667	0.494	2402	96	2231	2610

Warning: The estimated sample coverage(overlapping information)is too low so that Nhat is unstable. Recommend the use of Nhat-0 or Nhat-1.

Parameter estimates:

	u1	u2	u3	r12	r13	r23
Nhat-0	0.41	0.18	0.27	-0.32	-0.22	1.08
Nhat	0.25	0.11	0.17	0.10	0.25	2.36
Nhat-1	0.37	0.16	0.24	-0.24	-0.13	1.32

Definitions for the sample coverage approach:

M: number of individuals ascertained in at least one list.

D: the average of the number of invididuals listed in the combination of any two lists omitting the other one.

C[~]: sample coverage estimate, see Equation (14) of Chao and Tsay(1998).
est: population size estimate.

se: estimated standard error of the population size estimation based on bootstrap replications.

cil: 95% confidence interval lower limit(using a log-transformation).

ciu: 95% confidence interval upper limit(using a log-transformation).

Nhat: Population size estimate for sufficiently high sample coverage cases, see Equation (20) of Chao and Tsay (1998).

Nhat-1: One-step population size estimate for low sample coverage cases; see Equation (2.21) of Chao et al. (1996). This estimator is

suggested for use when the estimated se of Nhat is relatively large.
u1,u2,u3: estimated mean probabilities depending on the estimate of N.
r12,r13,r23 etc.: estimated coefficient of covariation(CCV) depending on the
estimate of N.

```
# Print the CARE1 input format

## n1 = reared = 886
## n2 = core = 388
## n3 = peripheral = 576

## Using Nhat-1

Nhat1 <- round(estN(micro_care, method = "LSC", se = TRUE, nboot = 200))

micro_all_estimate <- Nhat1[1]
micro_all_CI <- c(Nhat1[3], Nhat1[4])
```

Step 2. Estimating the undersampling ratio for Core Microgastrinae.

```
## proportion of core malaise of the ichao total

proportion_core <- nrow(core)/micro_all_estimate

proportion_core
```

```
[1] 0.1615321
```

```
CI_proportion_core <- nrow(core)/micro_all_CI

CI_proportion_core
```

```
[1] 0.1738351 0.1487730
```

Step 3 Calculate total number of insects in ACG

```
## Total number of insecta bins in ACG core malaise
n_insects_core <- nrow(incr)

n_insects_core
```

```
[1] 53945
```

```
## Total number of Braconidae bins in ACG core malaise
n_brac_core <- nrow(Brac)

n_brac_core
```

```
[1] 2160
```

```
## Total number of Ichneumonoidea bins in ACG core malaise
n_ichn_core <- nrow(Ichn)

n_ichn_core
```

```
[1] 3402
```

```
## Total number of hymenoptera bins in ACG core malaise
n_hym_core <- nrow(hym)

n_hym_core
```

```
[1] 13934
```

```
## insecta
total_insec_acg <- n_insects_core*(1/proportion_core)

total_insec_acg
```

```
[1] 333958.5
```

```
CI_total_insect_acg <- n_insects_core*(1/round(CI_proportion_core,3))

n_insects_core*(1/round(CI_proportion_core,3))
```

```
[1] 310028.7 362047.0
```

```
n_insects_core*(1/CI_proportion_core)
```

```
[1] 310322.8 362599.4
```

```
CI_total_insect_acg
```

```
[1] 310028.7 362047.0
```

```
## Braconidae  
total_brac_acg <- n_brac_core*(1/proportion_core)  
  
total_brac_acg
```

```
[1] 13371.96
```

```
CI_total_brac_acg <- n_brac_core*(1/round(CI_proportion_core,3))  
  
n_brac_core*(1/round(CI_proportion_core,3))
```

```
[1] 12413.79 14496.64
```

```
n_brac_core*(1/CI_proportion_core)
```

```
[1] 12425.57 14518.76
```

```
CI_total_brac_acg
```

```
[1] 12413.79 14496.64
```

```
## Ichneumonoidea  
total_ichn_acg <- n_ichn_core*(1/proportion_core)  
  
total_ichn_acg
```

```
[1] 21060.84
```

```
CI_total_ichn_acg <- n_ichn_core*(1/round(CI_proportion_core,3))
```

```
n_ichn_core*(1/round(CI_proportion_core,3))
```

```
[1] 19551.72 22832.21
```

```
n_ichn_core*(1/CI_proportion_core)
```

```
[1] 19570.27 22867.05
```

```
CI_total_ichn_acg
```

```
[1] 19551.72 22832.21
```

```
## Hymenoptera
```

```
total_hym_acg <- n_hym_core*(1/proportion_core)
```

```
total_hym_acg
```

```
[1] 86261.52
```

```
CI_total_hym_acg <- n_hym_core*(1/round(CI_proportion_core,3))
```

```
n_hym_core*(1/round(CI_proportion_core,3))
```

```
[1] 80080.46 93516.78
```

```
n_hym_core*(1/CI_proportion_core)
```

```
[1] 80156.41 93659.46
```

```
CI_total_hym_acg
```

```
[1] 80080.46 93516.78
```

```
## compile results

compiled_results <- data.frame(Taxonomic_group = c('Insecta', "Hymenoptera",
                                                  "Ichneumonoidea", "Braconidae"),
                              total_core = c(n_insects_core, n_hym_core,
                                              n_ichn_core, n_brac_core),
                              total_acg = c(total_insec_acg, total_hym_acg,
                                              total_ichn_acg, total_brac_acg),
                              CI_low = c(CI_total_insect_acg[1],
                                          CI_total_hym_acg[1],
                                          CI_total_ichn_acg[1],
                                          CI_total_brac_acg[1]),
                              CI_high = c(CI_total_insect_acg[2],
                                           CI_total_hym_acg[2],
                                           CI_total_ichn_acg[2],
                                           CI_total_brac_acg[2]))

compiled_results
```

	Taxonomic_group	total_core	total_acg	CI_low	CI_high
1	Insecta	53945	333958.48	310028.74	362046.98
2	Hymenoptera	13934	86261.52	80080.46	93516.78
3	Ichneumonoidea	3402	21060.84	19551.72	22832.21
4	Braconidae	2160	13371.96	12413.79	14496.64

Step 4. Estimating global insect richness – based on trees

```
##### Total insects

print("Total world insects based on estimated tree richness")
```

```
[1] "Total world insects based on estimated tree richness"
```

```
total_insect_world <- (TW*total_insec_acg)/TA
total_insect_world
```

```
[1] 16313649
```

```
CI_total_insect_world <- (TW*CI_total_insect_acg)/TA
```

```
CI_total_insect_world
```

```
[1] 15144697 17685754
```

```
##### Total Hymenoptera
```

```
print("Total world insects based on estimated tree richness")
```

```
[1] "Total world insects based on estimated tree richness"
```

```
total_hym_world <- (TW*total_hym_acg)/TA
```

```
total_hym_world
```

```
[1] 4213818
```

```
CI_total_hym_world <- (TW*CI_total_hym_acg)/TA
```

```
CI_total_hym_world
```

```
[1] 3911877 4568232
```

```
##### Total Ichneumonoidea
```

```
print("Total world insects based on estimated tree richness")
```

```
[1] "Total world insects based on estimated tree richness"
```

```
total_ichn_world <- (TW*total_ichn_acg)/TA
```

```
total_ichn_world
```

```
[1] 1028808
```

```
CI_total_ichn_world <- (TW*CI_total_ichn_acg)/TA
```

```
CI_total_ichn_world
```

```
[1] 955088.7 1115338.5
```

```
##### Total Braconidae
```

```
print("Total world insects based on estimated tree richness")
```

```
[1] "Total world insects based on estimated tree richness"
```

```
total_brac_world <- (TW*total_brac_acg)/TA
total_brac_world
```

```
[1] 653211.3
```

```
CI_total_brac_world <- (TW*CI_total_brac_acg)/TA
CI_total_brac_world
```

```
[1] 606405.5 708151.4
```

```
## compile results
```

```
compiled_results_all <- bind_cols(compiled_results,
                                   data.frame(total_word =
                                               c(total_insect_world,
                                                 total_hym_world,
                                                 total_ichn_world,
                                                 total_brac_world),
                                               CI_w_low = c(CI_total_insect_world[1],
                                                           CI_total_hym_world[1],
                                                           CI_total_ichn_world[1],
                                                           CI_total_brac_world[1]),
                                               CI_w_high = c(CI_total_insect_world[2],
                                                            CI_total_hym_world[2],
                                                            CI_total_ichn_world[2],
                                                            CI_total_brac_world[2])))

compiled_results_all
```

	Taxonomic_group	total_core	total_acg	CI_low	CI_high	total_word
1	Insecta	53945	333958.48	310028.74	362046.98	16313649.1
2	Hymenoptera	13934	86261.52	80080.46	93516.78	4213817.5
3	Ichneumonoidea	3402	21060.84	19551.72	22832.21	1028807.8
4	Braconidae	2160	13371.96	12413.79	14496.64	653211.3

	CI_w_low	CI_w_high
1	15144697.0	17685753.6
2	3911877.1	4568232.3
3	955088.7	1115338.5
4	606405.5	708151.4